

Artificial Intelligence

Introduction

Artificial Intelligence (AI) is the field of computer science focused on creating systems that can perform tasks that normally require human intelligence. These tasks include reasoning, learning, perception, language understanding, and decision making.

AI Applications

AI is used in virtual assistants, recommendation systems, fraud detection, medical image analysis, robotics, autonomous vehicles, and many other applications.

Machine Learning and AI

Machine learning is an important approach within AI. Instead of explicitly programming every rule, a machine learning system learns patterns from data and uses those patterns to make predictions or decisions.

Benefits and Challenges

AI can automate repetitive work, support decision making, and process large amounts of information. Challenges include data quality, bias, privacy, security, explainability, and the need for suitable computing resources.